

Flat-Top Pulse Generation Using 3-dB Abrupt Taper Interferometric Pulse Shaper

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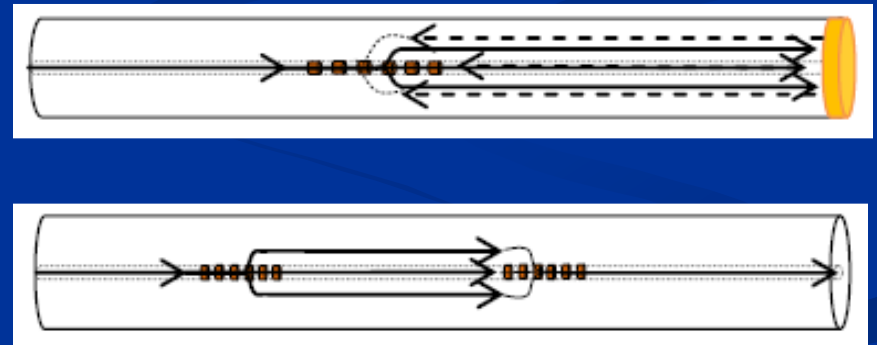
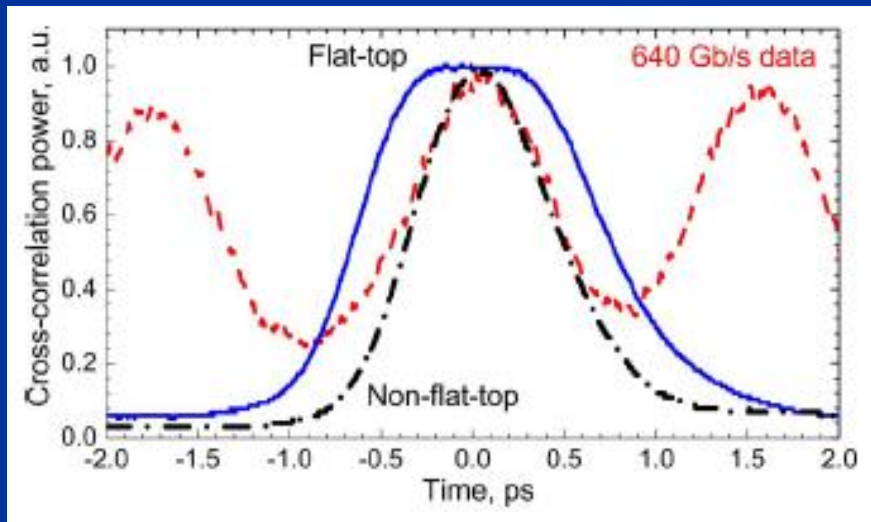


Outline

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Introduction

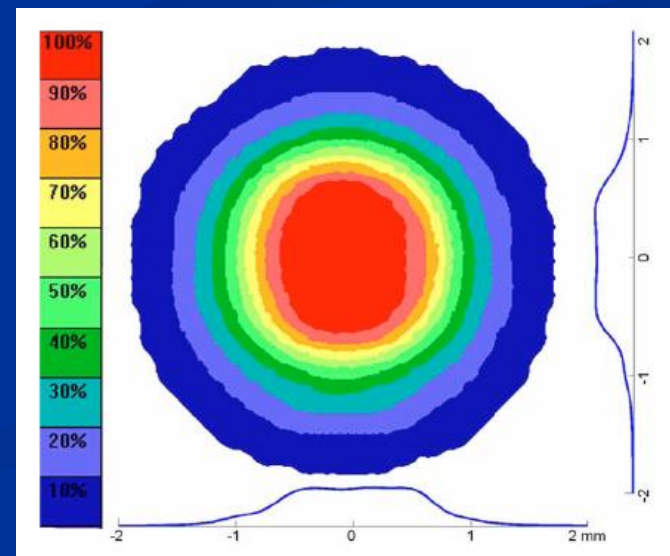
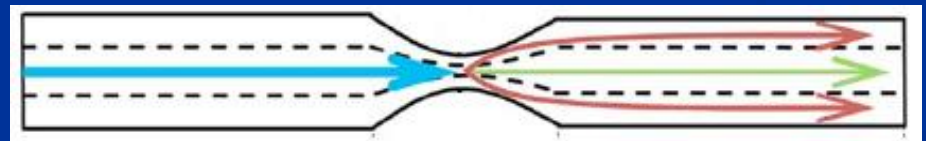
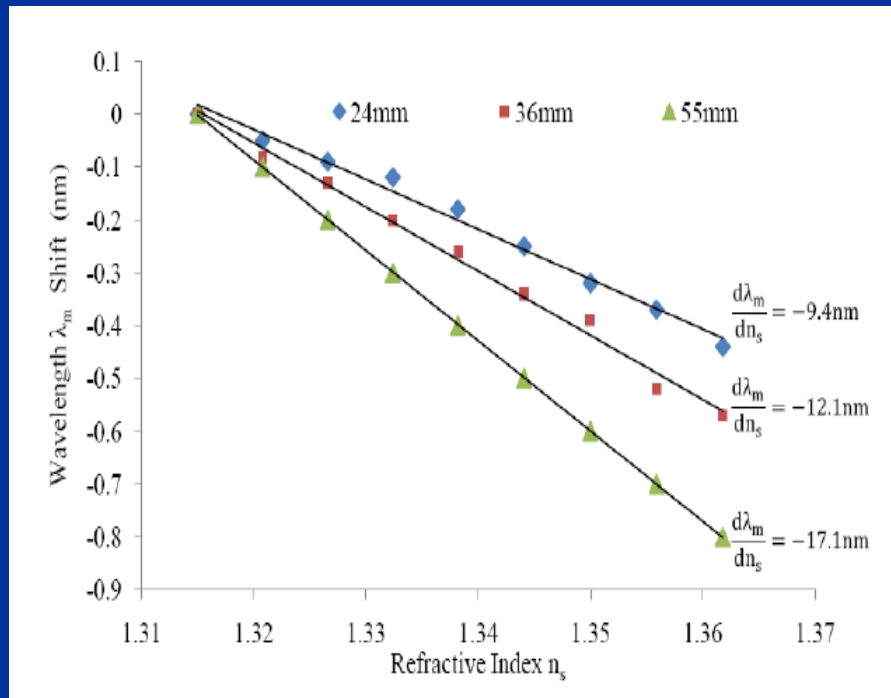
- Flat-top pulses:
 - Have useful high-speed telecom applications
 - Can be generated using non-linear polarization rotation, LPG interferometers, etc.



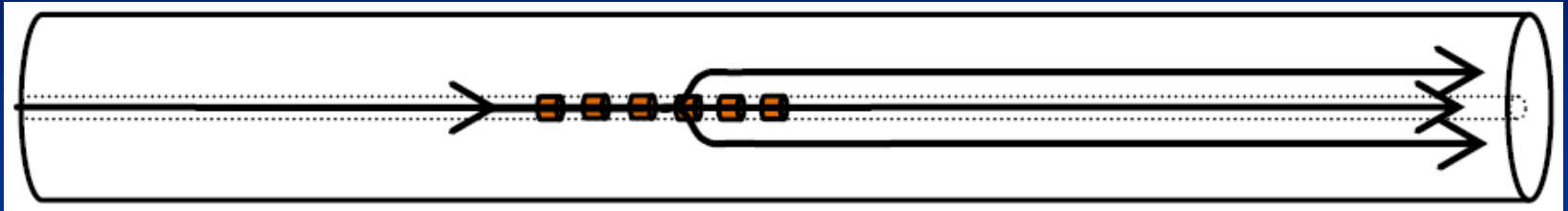
Oxenlowe et al. "640 Gb/s Timing Jitter-Tolerant Data Processing Using a Long-Period Fiber-Grating-Based Flat-Top Pulse Shaper." J. Sel. Top. Q. Elec., vol. 14, no. 3, May/June 2008. pp. 566 - 572.

Introduction (2)

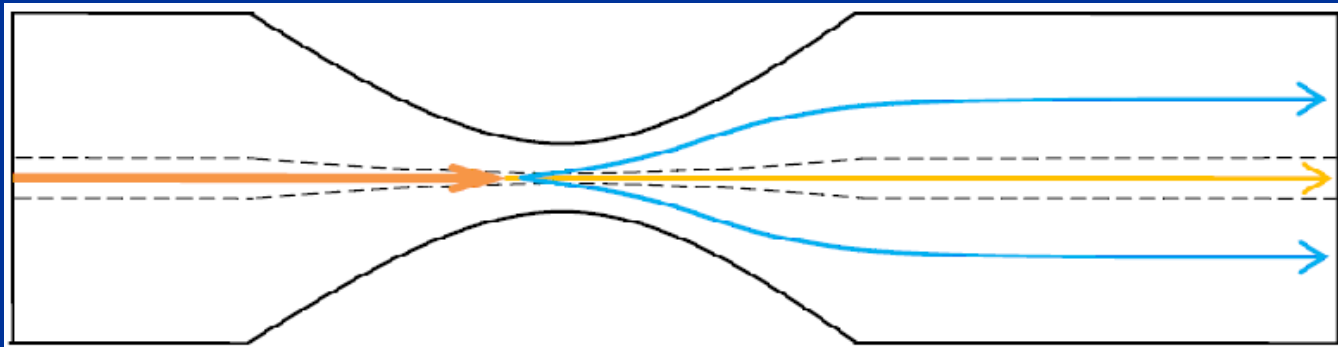
- Abrupt tapers:
 - Have recently been used similar to LPGs in interferometers for various sensor applications, beam shaping and the generation of flat-top pulses.



LPG vs. Abrupt Taper

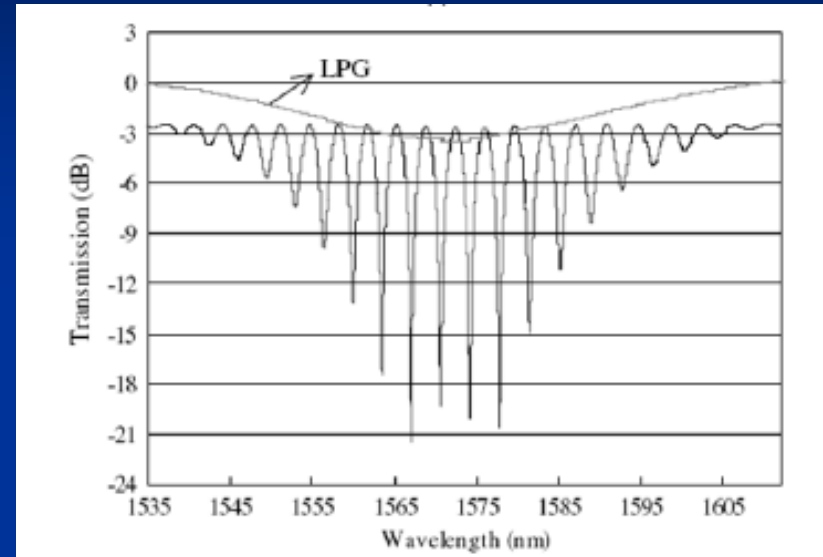
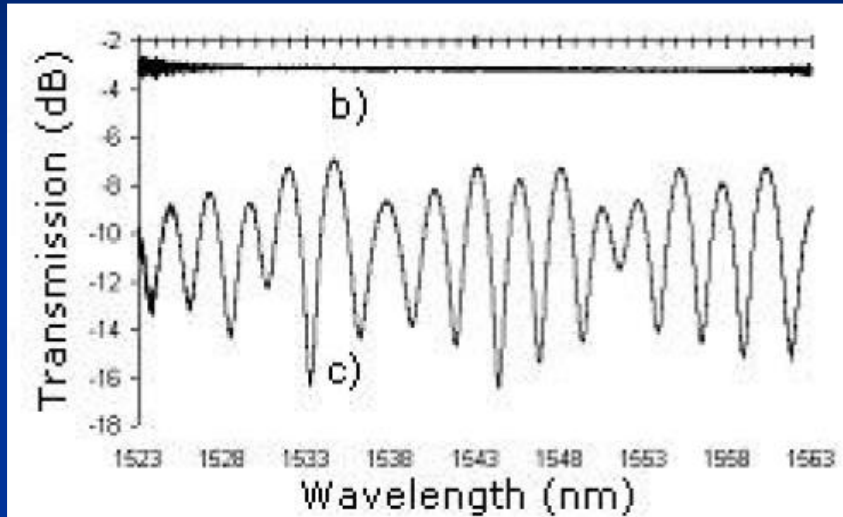


$$\lambda = (\beta_{co} - \beta_{cl}) \Lambda$$



- Principle of operation very similar
 - Coupling of power from the core mode to the various cladding modes
 - LPG coupling dependent on phase matching based on grating period, tapers do not have this dependence

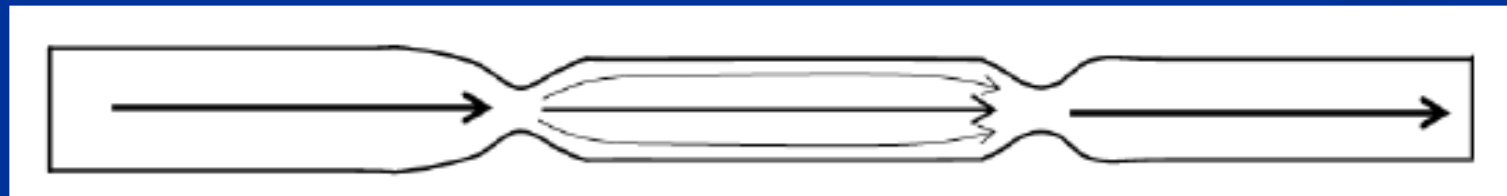
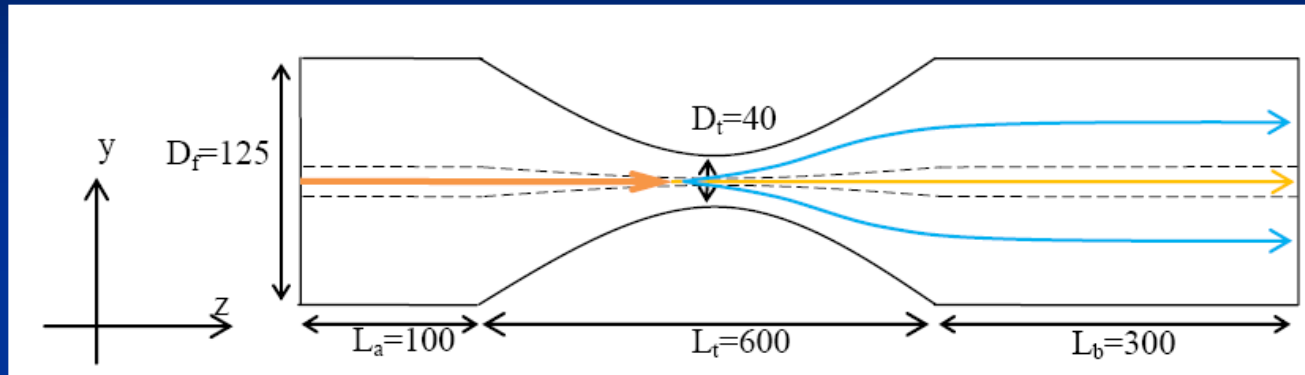
LPG vs. Abrupt Taper (2)



K. S. Chiang, Y. Liu, M. N. Ng, and X. Dong, "Analysis of etched long-period fibre grating and its response to external refractive index," *Electron. Lett.* 36, 966, (2000).

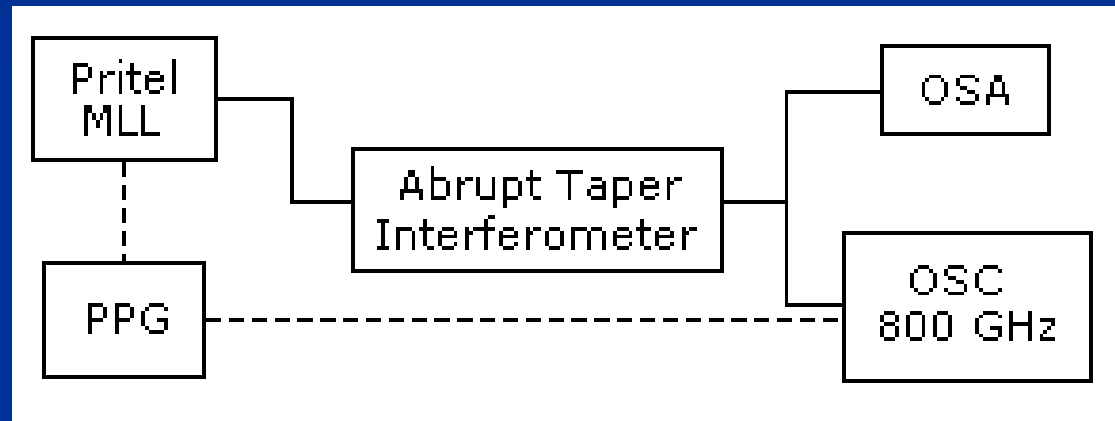
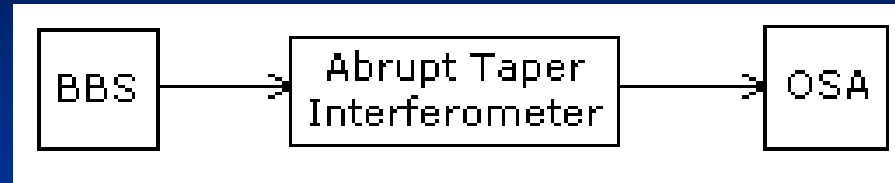
- **LPGs:**
 - High extinction ratio, wavelength dependent, high cost
- **Abrupt Taper:**
 - Lower extinction ratio, wavelength independent, low cost

Interferometer Fabrication



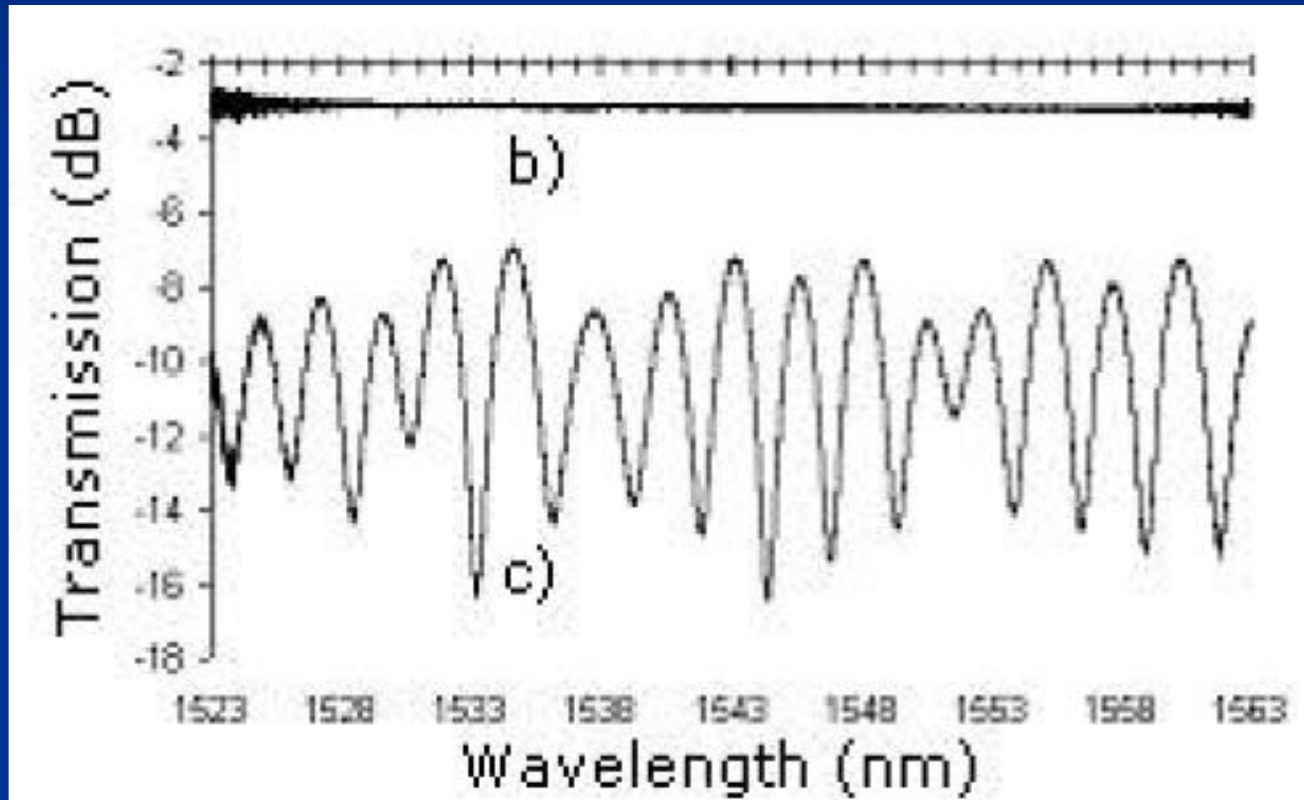
- Abrupt Taper Mach-Zehnder Interferometer:
 - Each taper with ~ 40 μm waist, 600 μm long.
 - Interferometer length: ~ 20 cm.
 - Necessary for proper filter bandwidth.

Experimental Setup

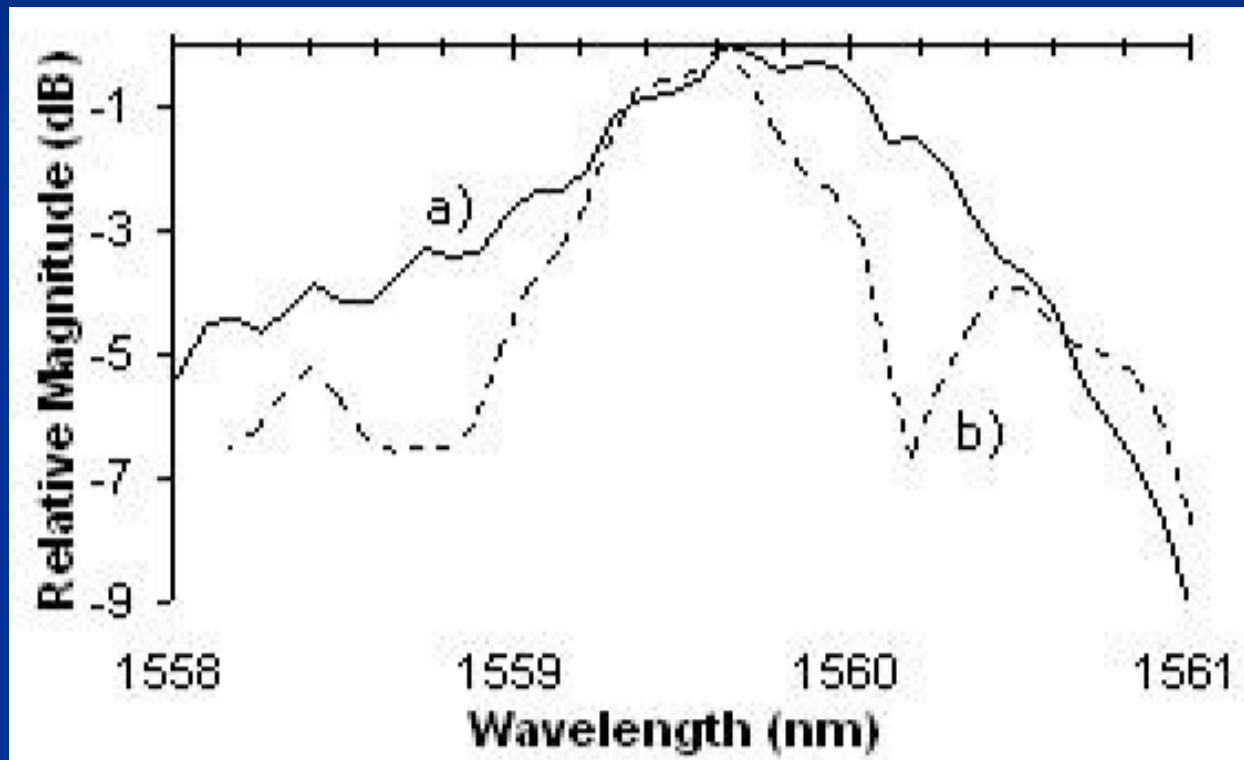


- Pritel Mode-Locked Laser
 - 2.2 ps Gaussian pulses, wavelength tunable.
- 800 GHz BW Agilent Ultra Wide-band Optical Sampler

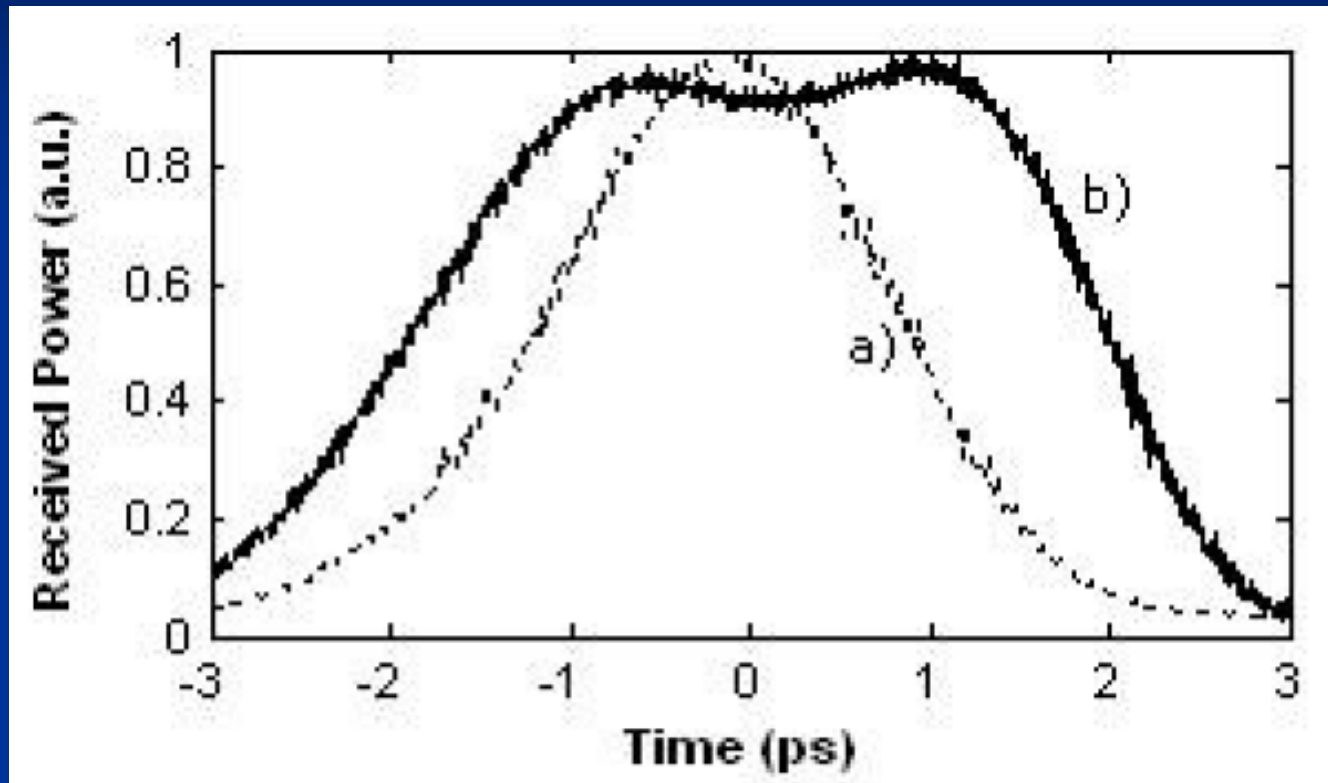
Experimental Results



Experimental Results (2)



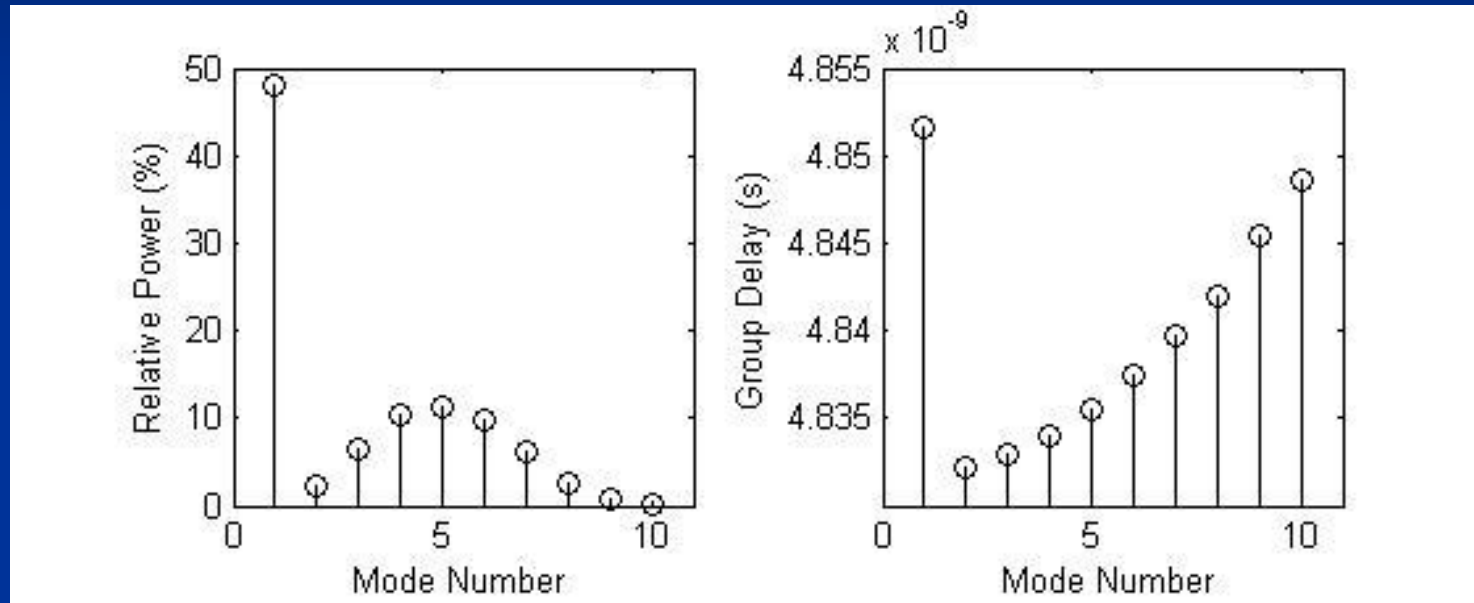
Experimental Results (3)



- 2.2 ps Gaussian input
- 3.8 ps flat output
- 5% ripple

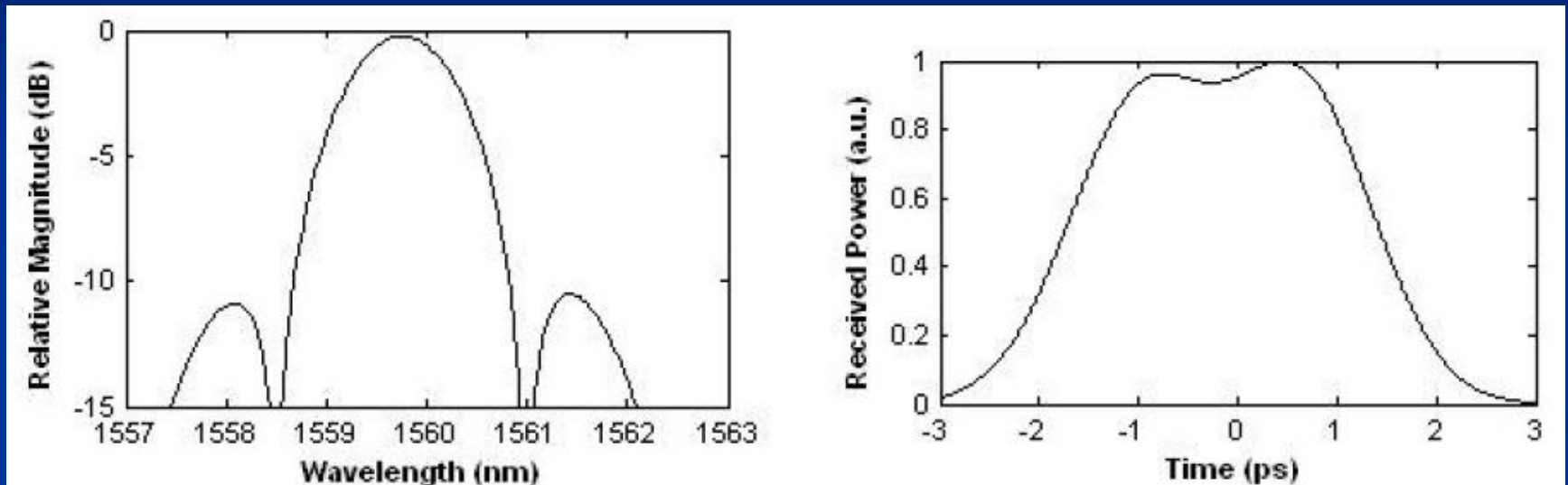
Simulation Theory

$$H(f) = \sum a_m \exp\{-b(f-f_0)^2\} \exp\{-j2\pi\tau_m L(f-f_0)\}$$



- Modeled as Multimode Fiber
- Relative mode power determined from BPM simulation of abrupt taper, 40 um waist, 600 um long.
- Direct simulation of SMF-28 to get group delays.

Simulation Results



- Simulated interferometer length of 20.1 cm
- Final 3.3 ps pulsewidth, 4% ripple
 - Experiment: 3.8 ps, 5% ripple
 - discrepancy due to assumption of transform limited input pulses and low extinction ratio of actual interferometer.

Summary and Conclusion

- Showed an inexpensive, wavelength independent, abrupt taper Mach-Zehnder interferometer pulse shaper
- Generates 3.8 ps flat-top pulses from 2.2 ps Gaussian input pulses with 5% ripple.

	Pulsewidth	Ripple	I/O Ratio	Cost
Long-Period Grating ¹	2 ps	4%	2.2:1	High
Abrupt Taper	3.8 ps	5%	1.73:1	low

¹R. Slavik et al., "Long-Period Fiber-Grating-Based Filter for Generation of Picosecond and Subpicosecond Transform-Limited Flat-Top Pulses." PTL, vol. 20, no. 10, May 2008, pp. 806 – 808.

Acknowledgements

- The experiment was conducted at the National Microelectronics and Photonics Testing Collaboratory in Kingston, ON.

